ANUSHKA VERMA

https://github.com/anushkav03 | linkedin.com/in/anushkav6
Berkeley, CA | (341) 766-8946 | anushka.v@berkeley.edu

EDUCATION

University of California, Berkeley December 2025
B.A. Major in Data Science, Minor in Computer Science

Relevant Coursework: Data Structures & Algorithms, Efficient Algorithms, Software Engineering, Internet Architecture, Programming in R, Quantitative Linguistics, Principles of Data Science, Artificial Intelligence, Machine Learning, Computer Architecture, Inference and Decisions

SKILLS

Languages: Java, Python, C, Ruby, SQL, HTML/CSS, JavaScript, R, RISC-V assembly
Tools & Frameworks: React, Ruby on Rails, FastAPI, Postman, Git, AWS Bedrock, Valgrind, GDB
Libraries: Regex, Pandas, NumPy, sklearn, Matplotlib, plotly, seaborn, Junit, FastAPI, Pydantic

WORK EXPERIENCE

UC Berkeley MELC Department (Aug 2024 – May 2025) | *Research Apprentice – 3D Book of the Dead Project*

- Contributed to a project for digitizing and publishing interactive 3D scans of ancient Egyptian coffins using photogrammetry, 3D rendering, and visualization
- Gained experience in end-to-end 3D asset processing and web integration, using Metashape for photogrammetry and reconstruction, and prototyping website with interactive 3D models (glTF format)
- Analyzed and documented a custom XML/XSLT pipeline that dynamically generates HTML pages integrating assets
- Debugged data inconsistencies and missing assets in the pipeline; created an internal reference mapping for mislabeled or incomplete 3D models and metadata.
- Built a website prototype (HTML/CSS, JavaScript, XML/XSLT); implemented accessibility enhancements (semantic markup, color contrast, keyboard navigation).

OpenProject – Client Project with LMNT AI (Aug 2024 – May 2025) | *Technical Lead at OpenProject (Software Development Club)*

- Developed a React-based website with educational games, to address lack of early-education resources in low-income communities
- Spearheaded creation of a spelling-bee game using **Pygame**, incorporating custom artwork and sound effects; improved player engagement by 45% based on user testing
- Conducted research to inform game design and optimize learning outcomes for primary school students
- Integrated LMNT AI's **text-to-speech API** into the platform, incorporating feedback from 5+ educators and delivering weekly updates
- Coordinated structured code reviews and testing to improve code quality and teamwork efficiency

UC Berkeley EECS Department (Jan 2022 – May 2023) | *Academic Intern*

- Guided bi-weekly programming lab sections for 30+ students, achieving a 91% average student satisfaction rating across all sections based on university feedback surveys
- Underwent weekly professional development sessions focused on pedagogical enhancement and computer science principles

Private Tutor (Aug 2022 – May 2023)

- Designed and conducted classes for 4 students learning computer science content
- Created lesson plans and tested understanding with weekly sessions; all students scored A/A- grade

HIGHLIGHTED PROJECTS

Speech Odyssey – Gamified Speech Therapy Tool (University of Washington DubHacks Sept 2024)

- Created a low-cost speech therapy practice platform, making it an affordable solution compared to traditional speech therapy
- Engineered a text based choose-your-own-adventure game, with dynamic story generation powered by Claude 3.5 Sonnet from AWS Bedrock
- Utilized OpenAI Whisper for speech transcription & analysis, WebSockets for real-time voice interaction
- *Tech Stack:* Python, JavaScript, AWS Bedrock, FastAPI, Uvicorn, OpenAI Whisper, WebSockets

Stress Penguin – Chronic Stress Support Tool (UC Berkeley AI Hackathon Jun 2024)

- Built a machine learning model using Random Forest Classifier, trained on 400k+ data points to detect stress based on metrics like body temperature, blood oxygen level, heart rate variability, respiratory rate, and REM sleep
- Implemented real-time stress detection from Apple Watch data, sending notifications upon detecting stress abnormalities
- Integrated with Hume AI's empathetic AI voice model to provide immediate support to users upon stress detection
- *Tech Stack*: Swift, Xcode, Python, Swift, Hume API, WebSockets

Weather Dashboard (June 2024)

- Developed a full-stack weather news dashboard that integrates external APIs (Weather, Spotify OAuth), backend routing in Flask, and responsive front-end components with Bootstrap
- Implemented dynamic playlist recommendations and animated visuals, building reusable UI components and API-driven features

- *Tech Stack*: HTML/CSS, JavaScript, Bootstrap, Python, Flask, Render, Gunicorn, OAuth, Spotify API, oEmbed

Baby Bytes – Smart Parenting Assistant (UC Berkeley CalHacks Oct 2024)

- Developed a tool for new parents to track & analyze baby's health data for metrics like height, weight, sleep, feeding patterns
- Integrated Deepgram for voice input, and Google Cloud Vision for analyzing food intake from before-and-after meal photos, allowing busy parents to log data hands-free
- Key features: data visualizations (e.g. daily and weekly patterns), insights into nutrition and taste preferences
- *Tech Stack*: Python, FastAPI, SingleStore, Google Cloud Vision API, Deepgram API

Build Your Own World (May 2023) *GitHub link available upon request*

- Designed and implemented a 2D tile-based game in Java; developed a world-generator algorithm which creates unique, randomly-generated "worlds"
- Key features: customizable user avatar and game environment/theme, keyboard input and interaction, save and reload capability, engaging game mechanics
- Collaborated in a team setting, adhering to tight deadlines and performance review-style assessments, similar to real-world software development life cycles